

Rahul Tammalla

AI Software Engineer

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SUMMARY

AI Software Engineer building production agentic LLM systems and RAG pipelines, along with the APIs and cloud backends on AWS that put them in front of real users. Strong software-engineering fundamentals in Python, FastAPI, Docker, and CI/CD, with a focus on reliable, low-latency products, backed by M.S.-level research on LLM evaluation and human preference modeling.

SKILLS

LLM & Generative AI: RAG, Multi-Agent Systems, Prompt Engineering, Fine-Tuning, LangChain, LangGraph, Hugging Face Transformers, PyTorch, Mistral, Gemini, Groq, Anthropic API, Structured Output, Hallucination Mitigation, Agentic Reasoning, LLM Evaluation
Software & API Engineering: Python, FastAPI, Django, Pydantic, REST APIs, Microservices, Async I/O, Git, Unit Testing, System Design
Retrieval & Data: FAISS, ChromaDB, pgvector, Semantic Search, Embeddings (GTE/BERT), SQL, PostgreSQL, Pandas, NumPy, PySpark, Airflow
Cloud, DevOps & MLOps: Docker, Kubernetes, AWS (S3, Glue, Lambda, EMR), CI/CD, MLflow, Model Monitoring, scikit-learn, XGBoost

EXPERIENCE

I4DI (Institute for Development Impact)

Washington, DC

AI Engineer Intern

May 2025 – Aug 2025

- Built a **multi-agent** survey-generation system with **Mistral 7B** and **Claude**, using structured prompting and **JSON-schema** validation to fully automate survey authoring workflows from brief to delivery, cutting manual authoring effort by **90%**.
- Designed a **RAG** pipeline with a real-time knowledge-base override that prioritized uploaded documents over base-model knowledge, returning context-scoped answers in under **3 seconds** at **95%+** relevance across all user query types.
- Optimized **GTE** embedding pipelines in **PyTorch** with batched inference and async data loading, cutting embedding latency by **42%** (5.1s → 2.9s) and lifting retrieval Precision@K by **30%** across the full production knowledge base of the system.
- Shipped **FAISS**-backed semantic search with hybrid retrieval and metadata filtering, delivering sub-second query response across all analytics, search, and real-time retrieval workloads in the production **multi-agent** application.

Clarkson University

Potsdam, NY

Machine Learning Research Assistant

Jan 2025 – Jul 2025

- Analyzed **20M+** NEMSIS emergency records with statistical modeling, hypothesis testing, and **ML**, engineering **70+** features to surface 15 significant predictors of stroke-related 911 delays projected to cut emergency response time by **40%**.
- Built scalable **Python/SQL** pipelines that resolved **1M+** missing values to **95%** data quality via multivariate imputation, then trained tuned **Random Forest**, **XGBoost**, **LightGBM**, **SVM** models that consistently lifted overall prediction accuracy by **20%**.
- Built an **NLP** extraction pipeline with **spaCy**, BeautifulSoup, and regex that structured unstructured biological research text into **300+** normalized entities, boosting downstream data quality by **45%** for statistical and NLP modeling workflows.

Cognizant Technology Solutions

Hyderabad, India

Data Engineer

Oct 2021 – Aug 2024

- Built **Airflow/SQL** ETL pipelines on **Snowflake/Redshift** that processed **5M+** records daily, lifting reliability by **70%** and query performance by **30%**, enabling real-time reporting and automated data delivery across all downstream business teams.
- Engineered cloud-native **AWS** (S3, Glue, Lambda, EMR) pipelines handling **1M+** daily transactions at **95%+** uptime via **CI/CD** on **Kubernetes**, then built **5+** Tableau/Power BI dashboards that cut overall manual reporting time by **30%** for all teams.

PROJECTS

Agentic AI – DevOps Agent Swarm | [Python, LangGraph, NATS JetStream, FastAPI, Docker, Gemini]

[Github](#)

- Built an autonomous **6-agent swarm** (Observer, Diagnoser, Remediator, Safety, Orchestrator, Learner) with **LLM-powered** root-cause diagnosis over **NATS JetStream**, executing incident remediation via an FSM lifecycle with human-in-the-loop safety gates.

NLP – Clinical Consent NER + Risk Detector | [Python, spaCy, Transformers]

[Github](#)

- Fine-tuned a Transformer-based **NER** model on clinical consent documents, reaching **85% F1** across 5 entity types and **80%+** confidence on consent-language detection to automate risk classification across all clinical consent form types at scale.

Reinforcement Learning – Q-Learning Maze Optimization | [Python, Q-Learning, Epsilon-Greedy, Policy Learning]

[Demo](#)

- Implemented a **Q-Learning** agent with epsilon-greedy exploration on dynamic maze environments, achieving **80%+** policy convergence through reward shaping, epsilon decay scheduling, and systematic grid-search hyperparameter tuning.

EDUCATION

Clarkson University

Potsdam, NY

M.S. Applied Data Science, Merit Scholar; GPA: 3.8/4.0 (Graduate TA, Probability & Statistics)

Aug 2024 – May 2026

Amrita Vishwa Vidyapeetham

Kerala, India

B.Tech, Computer Science Engineering

May 2017 – Jun 2021

RESEARCH

Modeling Human Preference Signals in LLM Pairwise Evaluations, Clarkson University, May 2026. [\[Link\]](#)